1. Experiment Planning (Discussions Summary)

Goal: Improve LLM response quality by refining system instructions. **Hypothesis:** If we replace the current system prompt (Variant A) with a structured chain-of-thought prompt (Variant C), then positive 4-5 satisfaction ratings will improve because the AI's responses will be clearer, more complete, and better aligned with user intent.

- **Variants:** * **A (Control):** The current system prompt.
    - **C (Treatment):** The new structured system prompt with hidden reasoning.
- **KPIs:** * *Primary:* Increase the percentage of 4-5 satisfaction ratings.
    - *Secondary:* Response time (measuring latency).
- **Randomization Strategy:** Users are randomly assigned to either group with a 50/50 split at the user level, triggered when they first interact with the AI assistant.
- **Stopping Conditions:** Run the test for at least 2 weeks, ensuring each variant (arm) has a minimum of 1,000 rated chats. The test will be stopped immediately if Variant C begins hallucinating or returning broken replies.
- **Assumptions & Risks:**
    - *Assumption:* The traffic split is truly random and independent.
    - *Risk:* Increased Latency. The "chain of thought" naturally demands more processing time, posing a speed trade-off.
    - *Risk Mitigation:* Implement a 14-day minimum test buffer duration to properly evaluate the impact of this slower response time. The team also noted that the new prompt style might "feel a bit weird at first."

2. Dataset Details

The test successfully ran until it exceeded the 1,000 minimum chat threshold per arm. The final dataset metrics were:

- **Variant A (Control):** * Visitors: 1,300
    - Positive Outcomes (4-5 stars): 767
    - Conversion / Satisfaction Rate: **59.00%**
- **Variant C (Structured):** * Visitors: 1,300
    - Positive Outcomes (4-5 stars): 899
    - Conversion / Satisfaction Rate: **69.15%**

3. Results Analysis

The comparison between the two variants clearly indicates a winner:

- **Uplift:** The structured prompt resulted in an **absolute uplift of +10.15 percentage points**, which equates to a **relative improvement of +17.21%** over the current prompt.
- **Statistical Significance:** The test reached a 95% Confidence Level with a Z-score of 5.40 and a p-value practically at zero (~0.000000). The confidence interval spanned from 6.49% to 13.82%.
- **Outcome:** Highly Statistically Significant. The probability that the observed improvement occurred by random chance is extremely low. Since the confidence interval is well above zero, the data strongly supports that Variant C outperforms Variant A.
- **Recommended Action: Roll Out Variant C.**